



Department of Education

**STEM Education
National Schools of Excellence
Technology Syllabus
Grade 12
Term 1**

Science

TECHNOLOGY

Engineering

Mathematics

Issued free to schools by the Department of Education

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SECRETARY'S MESSAGE

This Technology Syllabus is to be used exclusively by teachers of Technology to teach Grade 12 Students enrolled for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The Grade 11 STEM Technology Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Technology before undertaking the advanced concepts introduced in this syllabus.

This syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principal object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and Skills and apply them to solve every day problems whilst appreciating Technology as a fundamental field of science.

This STEM Technology Syllabus for Grade 12 constitutes six out of the ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Technology.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Technology is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Technology to human lives, including good health and environment.

In order to ensure that teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Technology to be used for the Grade 12 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Technology is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Technology help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Technology is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Technology.

The Grade 12 Technology Syllabus entails eight (8) Modules. Over time different Modules will be introduced.

Since Technology is a highly specialized subject students undertaking Technology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Technology through the process of imparting basic Technology knowledge and skills to students.

MODULE	TOPICS
12.1: JavaScript	1. JavaScript (Basics) 2. JavaScript Objects 3. JavaScript (Advance)
12.2: Software Development	1. Software Development Life Cycle (SDLC) 2. Programming Paradigms – PP vs. OOP
12.3: Python	1. Basic Python 2. Advance Python
12.4: Java	1. Basic Java 2. Java Object-Oriented
12.5: C Programming	1. Basic C Programming
12.6: C++ Programming	1. Basic C++ 2. Object Oriented C++ 3. Advance C++
12.7 Artificial Intelligence	1. Introduction to Artificial Intelligence (AI) 2. Mathematics of Artificial Intelligence (AI) 3. Science of Artificial Intelligence (AI) 4. Machine Learning
12.8 Robotics	1. Introduction to Robotics 2. Arduino Hardware 3. Arduino Software – Basics of Arduino IDE 4. MATLAB

Since Computer Science of Technology is a highly specialized subject students undertaking Technology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language Skills.

The modules in this Syllabus are sequenced to guide teachers of Technology through the process of imparting basic Technology knowledge and Skills to students.

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In the STEM Grade 11 Technology Syllabus, students were introduced to the basic concepts and knowledge of Technology in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Technology as they are applied to six Thema.

The introduction of the STEM Curriculum will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

NEW NOMENCLATURE

As part of the overall Curriculum Development, new nomenclature was introduced. These are explained below:

Term	Explanation
Domain	Each STEM area is defined as <i>domain</i> (e.g., Science Domain, Technology Domain, etc.)
Phylum	What would normally be regarded as “subject” (e.g., Technology, Physics, Technology, etc.) is now regarded as <i>phylum</i> ; the plural is <i>phyla</i> .
Thema	Thematic areas of studies
Module	A specific concept within the Thema which is being studied
Topic	A specific aspect of the Module that is being studied

STEM Domains

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, Technology and Technology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Technology, Technology and Physics.

Technology

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM GRADE 12 TECHNOLOGY CONTENT OVERVIEW – GRADE 12, TERM 1

Table 1.1 Sequencing & Benchmarking STEM Technology Content Standards

Setting STEM Technology Content Standards for Progressive Learning & Assessment			
Module 12.1: JavaScript	Module Content Standards		
	12.1: Utilizing JavaScript basic-to-advanced concepts on Web Development applications.		
Topics	Performance Indicators	Module Content Benchmark	Assessment Tasks
1. JavaScript (Basic)	12.1.1a: Enabling and disabling JavaScript support in browsers. 12.1.1b: Placing JavaScript in HTML files in different ways. 12.1.1c: Utilizing and/or manipulate the different JavaScript data types to support a software program. 12.1.1d: Declare and utilize variables with JavaScript. 12.1.1e: Write decision-making code using if...else...elseif conditional statements in JavaScript. 12.1.1f: Use switch...case statement to test or evaluate an expression with different values in JavaScript. 12.1.1g: Manipulate or perform the operations on variables and values using the operators in JavaScript. 12.1.1h: Coding loop statements to reduce the number of action lines. 12.1.1i: Utilizing break and continue statements to immediately exit a loop and begin another iteration of any loop respectively.	Students' ability to master the knowledge and skills of: Creating responsive and interactive elements of a Webpage/Website using programming skills of HTML, CSS and JavaScript.	Assignment + Test + Quiz

	12.1.1j: Using function codes to avoid writing the same codes. 12.1.1k: Coding and handling events. 12.1.1l: Code cookies into a webpage. 12.1.1m: Creating dialog boxes for a webpage. 12.1.1n: Implement the print function window object on the webpage.		
2. JavaScript Objects	1.2a: Create and use the number, boolean, string, array, date, math, and RegExp objects using JavaScript in the webpage 12.1.2b: Selecting and Styling DOM elements in JavaScript 12.1.2c: Get, set and remove attributes from HTML elements using JavaScript 12.1.2d: Manipulating HTML elements using JavaScript 12.1.2e: Navigate between DOM nodes using JavaScript 12.1.2f: Implementing and manipulating the JavaScript Window, Screen, Location, Window History, and Window Navigation Objects 12.1.2g: Using a timer to execute a function at a particular time.		
3. JavaScript (Advance)	12.1.3a: Using JavaScript date and time objects in webpages to get the user's local date and time by accessing the system clock through the browser. 12.1.3b: Perform mathematical operations using JavaScript.		

	12.1.3c: Convert the data type of different values using JavaScript. 12.1.3d: Using DOM event listeners using JavaScript. 12.1.3e: Utilizing the event propagation. 12.1.3f: Borrow functionality from existing objects in JavaScript. 12.1.3g: Able to gracefully deal with errors. 12.1.3h: Validating HTML forms using JavaScript. 12.1.3i: Creating animation on your webpage using JavaScript. 12.1.3j: Include multimedia on your webpage using JavaScript. 12.1.3k: Find and fix bugs with JavaScript. 12.1.3l: Creating a client-side map.		
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Module 12.2 Software Engineering	Module Content Standards		
	12.2a: Use the Software Development life Cycle and Models on a school software project. 12.2b: Organize codes using OOP and/or PP.		
Topics	Performance Indicators Students are expected to:	Module Content Benchmark	Assessment Tasks
1. Software Development Life Cycle (SDLC)	12.2.1a: Use each phase of the Software Development Life Cycle to tackle a software development project' 12.2.1b: Use software development idea to plan to solve a problem 12.2.1c: Appropriately apply the SDLC and a model to a software development scenario	Students' ability to master the knowledge and skills of: - The Software Development Life Cycle and how to use it design a Software; and - Organizing codes using Procedural Programming or Object-Oriented Programming.	Test + Case Study
2. Programming Paradigms - PP vs. OOP	12.2.2a: Differentiate between Procedural Programming (PP) and Object-Oriented Programming (OOP) 12.2.2a: Organizing codes using PP or OOP		

Module 12.3 Python	Module Content Standards		
	12.3: Utilizing Python basic-to-advanced concepts on Software Development applications such a mobile or pc game.		
Topics	Performance Indicators Students are expected to:	Module Content Benchmark	Assessment Tasks
1. Basic Python	12.3.1a: Set up the suitable environment to use Python programming 12.3.1b: Create variables using Python 12.3.1c: Use different Python operators 12.3.1d: Use algorithm knowledge with different decision-making statements using Python	Students' ability to master the knowledge and skills of: Basic and Advance Python concepts	Assignment Test + Quiz

	12.3.1e: Use algorithm knowledge with loop statements using Python 12.3.1f: Create strings using Python 12.3.1g: Construct list data types with Python 12.3.1h: Create tuples using Python 12.3.1i: Access dictionary elements using keys 12.3.1j: Manipulating date and time elements with Python 12.3.1k: Create a function using Python 12.3.1l: Produce I/O files using Python 12.3.1m: Use exception handling and/or assertion to handle any unexpected error in a Python program.	• Building a software application	
2. Advance Python	12.3.2a: Create a class of objects using Python 12.3.2b: Creating Instance Objects 12.3.2c: Accessing object attributes 12.3.2d: Destroying Objects 12.3.2e: Creating a CGI program 12.3.2f: Creating a database and interfacing your program to the database 12.3.2g: Utilize socket programming using Python 12.3.2h: Creating a multithread program 12.3.2i: Build a software application for your school		

Table 1.2 Mapping Content Progressions by Modules

Module	Content Standard	12.1	12.2	12.3
12.1 JavaScript	12.1: Utilizing JavaScript basic-to-advanced concepts on Web Development applications	✓		
12.2 Software Development	12.2a: Use the Software Development life Cycle and Models on a school software project. 12.2b: Organize codes using OOP and/or PP.		✓	
12.3 Python	12.3: Utilizing Python basic-to-advanced concepts on Software Development applications such a mobile or pc game.			✓

MODULE 12.1: JAVASCRIPT

3 Weeks

HTML, CSS and JavaScript are three main programming languages for Web Development. After utilizing HTML and CSS to build and style the webpage/website in a static manner, JavaScript is added to provide more dynamics to the web application.

This module aims to give students the basic-to-advance concepts of coding using the JavaScript language. The application surrounding this module is on web development, however, students can be challenged upon teacher's instruction to utilize the programming language for different applications.

The outline of the module is structured in the following manner:

Topic 1: JavaScript (Basic)

Topic 2: JavaScript Objects

Topic 3: JavaScript (Advance)

Linking Modules

Prerequisites:

- STEM-proper
 - Module 11.5 Computer Programming
 - Topic 2: HTML Basics
 - Topic 3: HTML Advance and HTML5 Features
 - Module 11.6 Web Development – Cascading Style Sheets (CSS)

Module Content Standard

12.1: Utilizing JavaScript basic-to-advanced concepts on Web Development applications

Module Content Benchmark

Students' ability to master the knowledge and Skills of:

- Creating responsive and interactive elements of a Webpage/Website using programming Skills of HTML, CSS and JavaScript.

Topic 1: JavaScript (Basic)

Performance Indicators

Students are expected to:

- 12.1.1a:** Enabling and disabling JavaScript support in browsers.
- 12.1.1b:** Placing JavaScript in HTML files in different ways.
- 12.1.1c:** Utilizing and/or manipulate the different JavaScript data types to support a software program.
- 12.1.1d:** Declare and utilize variables with JavaScript.
- 12.1.1e:** Write decision-making code using if...else...elseif conditional statements in JavaScript.
- 12.1.1f:** Use switch...case statement to test or evaluate an expression with different values in JavaScript.
- 12.1.1g:** Manipulate or perform the operations on variables and values using the operators in JavaScript.
- 12.1.1h:** Coding loop statements to reduce the number of action lines.
- 12.1.1i:** Utilizing break and continue statements to immediately exit a loop and begin another iteration of any loop respectively.
- 12.1.1j:** Using function codes to avoid writing the same codes.
- 12.1.1k:** Coding and handling events.
- 12.1.1l:** Code cookies into a webpage.
- 12.1.1m:** Creating dialog boxes for a webpage.
- 12.1.1n:** Implement the print function window object on the webpage.

Content Expansion

Body of Knowledge for Understanding

- *Introduction: What is JavaScript?*
 - JavaScript is a Client-Side Scripting Language
 - Example Mechanism
 - Server-Side Programming
 - Client-Side Programming
 - JavaScript is an Object-based Language
 - Encapsulation
 - Abstraction
 - Inheritance
 - Polymorphism
 - JavaScript is NOT an OOP
 - JavaScript is an Interpreter Language, not a Compiler Language
 - What is a Compiler?
 - What is an Interpreter?
 - Key Differences
 - JavaScript Features
 - Applications of JavaScript
- Source Code Editors apart from Notepad.
 - Sublime Text
 - Visual Studio Code
 - Downloading Visual Studios

- Installing Visual Studios Code
 - Windows
 - Macintosh
 - Installing Live Server Extension on Visual Studio Code
- JavaScript Syntax
 - First JavaScript Coding
 - Case Sensitivity
 - Identifiers
 - White Spaces and Line Breaks
 - Semicolons
- JavaScript Comments
 - Single Line Comments
 - Multi Line Comments
- JavaScript Statements
- JavaScript Expressions
- JavaScript Keywords and Reserved Words
- Enabling JavaScript in Web-browsers
 - Internet Explorer
 - Firefox
 - Chrome
 - Opera
 - Use a Warning for Non-JavaScript Browsers
- Placement of JavaScript Codes
 - Inside the <head> Section of the HTML File
 - Inside the <body> Section of the HTML File
 - Inside the <body> and <head> Sections
 - Externally Embed the JavaScript Code into the HTML File
- JavaScript Variables
 - var Declaration Method
 - Identifiers
 - Assignment Operator
 - Data Types
 - Creating Variables
 - One Statement, Many Variables
 - Value=Undefined
 - Redeclaring Variables
 - Arithmetic with JavaScript
 - JavaScript Dollar Sign \$
 - JavaScript Underscore (_)
 - Let Declaration Method
 - Let Cannot be Redeclared
 - Block Scope in JavaScript
 - Redeclaring Variables using var Keyword can cause issues
 - Hoisting Variables
 - When to use Const
 - Const Objects and Arrays
 - Const Arrays

- Constant Objects
 - Block Scope
 - Redeclaring var variable
 - Hoisting Const
- JavaScript Data Types
 - Primitive Data Types
 - Non-Primitive Data Types
- JavaScript Operators
 - Arithmetic Operators
 - Adding
 - Subtracting
 - Multiplying
 - Dividing
 - Remainder
 - Incrementing
 - Decrementing
 - Exponential
 - Comparison (Relational) Operators
 - Bitwise Operators
 - Logical Operators
 - Assignment Operators
 - Special Operators
- JavaScript If Statements
 - If Statement
 - If...Else Statement
 - If...Else...If Statement
- JavaScript Switch Statements
- JavaScript Loops
 - For Loop
 - While Loop
 - Do-While Loop
 - For-In Loop
- JavaScript Functions
 - Function Syntax
 - Function Arguments
 - Function with Return Value
 - Function Object
 - Function Methods
 - JavaScript Function call() Method
 - JavaScript Function toString() Method
 - Function Object Examples

Set of Skills for Proficiency Levels

- Proficiency with Programming languages
- Mathematical Skills
- Communication Skills
- Learning concepts and applying them to problems

- Problem Solving Capability

Traits of Attitudes and Values to govern their lives

- Determination to study more on concepts that are not easily understood
- Inquisitiveness
- Self-motivation

Topic 2: JavaScript Objects

Performance Indicators

Students are expected to:

- 12.1.2a:** Create and use the number, boolean, string, array, date, math, and RegExp objects using JavaScript in the webpage.
- 12.1.2b:** Selecting and Styling DOM elements in JavaScript.
- 12.1.2c:** Get, set and remove attributes from HTML elements using JavaScript.
- 12.1.2d:** Manipulating HTML elements using JavaScript.
- 12.1.2e:** Navigate between DOM nodes using JavaScript.
- 12.1.2f:** Implementing and manipulating the JavaScript Window, Screen, Location, Window History, and Window Navigation Objects.
- 12.1.2g:** Using a timer to execute a function at a particular time.

Content Expansion

Body of Knowledge for Understanding

- Creating JavaScript Objects
 - Object Creating via Object Literal
 - Creating Instance of Object Directly (Using New Keyword)
 - Creating Objects by using an Object Constructor
 - Define Method in JavaScript Object
 - Object Methods for JavaScript
- JavaScript Array
 - By Array Literal
 - By Creating Instance of Array Directly (Using a New Keyword)
 - By Using an Array Constructor (Using New Keyword)
 - Array Methods of JavaScript
- JavaScript String
 - Via String Literal
 - Via String Object (Using New Keyword)
 - String Methods for JavaScript
 - String charAt(index) Method
 - String concat(str) Method
 - String indexOf(str) Method
 - String lastIndexOf(str) Method
 - String toLowerCase() Method
 - String toUpperCase() Method
 - String slice(beginIndex, endIndex) Method
 - String trim() Method
 - String split() Method
- JavaScript Date
 - Construct
 - Date Methods
 - Example: Print Date Object
 - Example: Current Time
 - Example: Display Digital Clock
- JavaScript Math

- Maths Methods of JavaScript
- Math.sqrt(n)
- Math.random()
- Math.pow(m,n)
- Math.floor(n)
- Math.ciel(n)
- Math.round(n)
- Math.abs(n)
- JavaScript Number
 - Number Constants
 - Number Methods
- JavaScript Boolean
 - An Example of Boolean Script
 - Boolean Properties in JavaScript
 - Boolean Methods in JavaScript
- JavaScript Browser Object Model (BOM)
 - Window Object
 - Window Object methods
 - Example of JavaScript alert()
 - Example of JavaScript confirm()
 - Example of JavaScript prompt()
 - Example of JavaScript open()
 - Example of JavaScript setTimeout()
 - Navigator Object
 - Navigator Object Properties
 - Navigator Object Methods
 - Screen Object
 - Properties of Screen Objects
 - History Object
 - Location Object
- DOM with JavaScript
 - DOM Properties
 - DOM Methods
 - Accessing field value by DOM
 - getElementById() Method
 - getElementsByClassName() Method
 - getElementsByName() Method
 - getElementsByTagName() Method
 - JavaScript innerHTML property
 - JavaScript InnerText property
 - Nodes
 - Selectors
 - Styling
 - Get set attributes
 - Manipulation
 - Navigation

Set of Skills for Proficiency Levels

- Proficiency with Programming languages
- Mathematical Skills
- Communication Skills
- Learning concepts and applying them to problems
- Problem Solving Capability

Traits of Attitudes and Values to govern their lives

- Determination to study more on concepts that are not easily understood.
- Inquisitiveness
- Self-motivation

Topic 3: JavaScript (Advance)

Performance Indicators

Students are expected to:

- 12.1.3a:** Using JavaScript date and time objects in webpages to get the user's local date and time by accessing the system clock through the browser.
- 12.1.3b:** Perform mathematical operations using JavaScript.
- 12.1.3c:** Convert the data type of different values using JavaScript.
- 12.1.3d:** Using DOM event listeners using JavaScript.
- 12.1.3e:** Utilizing the event propagation.
- 12.1.3f:** Borrow functionality from existing objects in JavaScript.
- 12.1.3g:** Able to gracefully deal with errors.
- 12.1.3h:** Validating HTML forms using JavaScript.
- 12.1.3i:** Creating animation on your webpage using JavaScript.
- 12.1.3j:** Include multimedia on your webpage using JavaScript.
- 12.1.3k:** Find and fix bugs with JavaScript.
- 12.1.3l:** Creating a client-side image map.

Content Expansion

Body of Knowledge for Understanding

- JavaScript Error Handling
 - Syntax Errors
 - Runtime Errors
 - Logical Errors
 - The try...catch...finally statement
 - The throw statement
 - The onerror() method
- JavaScript Form Validation
 - Example
 - Basic Form Validation
 - Data Format Validation
- JavaScript to Create Animation
 - Manual Animation
 - Automated Animation
 - Rollover with a Mouse Event
- JavaScript to Include Multimedia
 - Checking for Plug-Ins
 - Controlling Multimedia
- Debugging Mistakes in Code using JavaScript
 - Error Messages in IE
 - Error Messages in Firefox or Mozilla
 - Error Notification
 - How to Debug a Script
 - Use a JavaScript Validator
 - Add Debugging Code to your programs
 - Use a JavaScript Debugger
 - Useful Tips

- Creating Client-side Image Map with JavaScript
- Using JavaScript for Browser Detection.
 - Navigator Properties
 - Navigator Methods
 - Browser Detection

Set of Skills for Proficiency Levels

- Proficiency with Programming languages
- Mathematical Skills
- Communication Skills
- Learning concepts and applying them to problems
- Problem Solving Capability

Traits of Attitudes and Values to govern their lives

- Determination to study more on concepts that are not easily understood
- Inquisitiveness
- Self-motivation

MODULE 12.2: SOFTWARE DEVELOPMENT

2-3 Weeks

This module introduces students to the process of software development. It encompasses knowledge around the stages of developing a software, types of software development and the possible application of software development to solve problems. Also, included is the two fundamental programming paradigms – Procedural Programming and Object-Oriented Programming.

The SDLC is an acronym for Software Development Life Cycle. It is a sequence of scheduled actions for DEVELOPING or altering Software Products. This Module will provide you an understanding of the SDLC fundamentals, as well as the many SDLC models and their applications. In addition, other related techniques such as Agile, Rapid Application Development, and Prototyping will also be covered.¹

Programming paradigms refer to a variety of distinct approaches to the programming process. Different paradigms reflect fundamentally different ways to utilizing programming to provide answers to certain sorts of issues. The majority of programming languages belong to one paradigm, although others include components from many paradigms. The procedural and object-oriented paradigms are two of the most important programming paradigms.²

The outline of the module is structured in the following manner:

Topic 1: Software Development Life Cycle (SDLC)

Topic 2: PP vs OOP

Linking Modules

Prerequisites: NIL

Module Content Standard

12.2a: Use the Software Development life Cycle and Models on a school software project

12.2b: Organize codes using OOP and/or PP

Module Content Benchmark

Students' ability to master the knowledge and Skills of:

- The Software Development Life Cycle and how to use it design a Software; and
- Organizing codes using Procedural Programming or Object-Oriented Programming

¹ [SDLC Tutorial \(tutorialspoint.com\)](https://www.tutorialspoint.com/sdgc/sdgc.htm)

² [Object-Oriented Programming vs. Procedural Programming - Video & Lesson Transcript | Study.com](https://www.study.com/academy/course/object-oriented-programming-vs-procedural-programming-video-lesson-transcript)

Topic 1: Software Development Life Cycle (SDLC)

Performance Indicators

Students are expected to:

- 12.2.1a:** Use each phase of the Software Development Life Cycle to tackle a software development project
- 12.2.1b:** Use software development idea to plan to solve a problem
- 12.2.1c:** Appropriately apply the SDLC and a model to a software development scenario

Content Expansion

Body of Knowledge for Understanding

- Overview
 - Why do we need to learn Software Development Basics?
 - Applications
- Introduction to Software Development Life Cycle (SDLC)
 - What is SDLC and what are its Stages?
 - Stage 1: Planning and Requirement Analysis
 - Stage 2: Defining Requirements
 - Stage 3: Designing the Product Architecture.
 - Stage 4: Building (Coding) the Product
 - Stage 5: Testing the Product
 - Stage 6: Deployment in the Market and Maintenance
- Models of SDLF
 - Waterfall Model
 - Iterative Model
 - Spiral Model
 - V Model
 - Big Bang Model
 - Agile Model
 - RAD Model
 - Software Prototype
- Types of Software Developments³
 - Frontend Development
 - Backend Development
 - Full-stack Development
 - Desktop Development
 - Web Development
 - Database Development
 - Mobile Development
 - Cloud Computing
 - DevOps Engineering
 - Security Engineering
- Types of Software Platforms
 - Utility Platform

³ [Types of Software Development - Software Testing \(gcreddy.com\)](https://www.gcreddy.com/types-of-software-development-software-testing/)

- Technology Platform
- Computing Platform
- Data Harvesting Platform
- Content Distribution Platform
- On-demand Service Platform

Set of Skills for Proficiency Levels

- Proficiency with Programming languages
- Mathematical Skills
- Communication Skills
- Learning concepts and applying them to problems
- Problem Solving Capability

Traits of Attitudes and Values to govern their lives

- Determination to study more on concepts that are not easily understood
- Inquisitiveness
- Self-motivation

Topic 2: Programming Paradigms – PP vs. OOP

Performance Indicators

Students are expected to:

- 12.2.2a:** Differentiate between Procedural Programming (PP) and Object-Oriented Programming (OOP).
- 12.2.2a:** Organizing codes using PP or OOP.

Content Expansion

Body of Knowledge for Understanding

- Introduction to Programming Paradigms
 - Procedural Programming (PP)
 - Object-Oriented Programming (OOP)

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MODULE 12.3: PYTHON

2-3 Weeks

Python is a general-purpose programming language, which means it can be used to create almost anything with the appropriate tools and libraries. It is ideal for backend web development, data analysis, artificial intelligence, and scientific computing in the professional world. It has also been used to create productivity tools, games, and desktop programs, so there are lots of resources to assist you learn how to do so.

This module aims to give students the basic-to-advance concepts of coding using the Python programming language. The application surrounding this module is on software development, particularly a simple app or video game. However, students can be challenged upon teacher's instruction to utilize the programming language for different applications.

The outline of the module is structured in the following manner:

Topic 1: Basic Python

Topic 2: Advance Python

Linking Modules

Link:

- STEM-proper
 - Module 11.4 Algorithms and Data Structures
 - Module 12.2 Software Development

Module Content Standard

12.3: Utilizing Python basic-to-advanced concepts on Software Development applications such a mobile or pc game.

Module Content Benchmark

Students' ability to master the knowledge and Skills of:

- Basic and Advance Python concepts
- Building a software application

Topic 1: Basic Python

Performance Indicators

Students are expected to:

- 12.4.1a:** Set up the suitable environment to use Python programming.
- 12.4.1b:** Create variables using Python.
- 12.4.1c:** Use different Python operators.
- 12.4.1d:** Use algorithm knowledge with different decision-making statements using Python.
- 12.4.1e:** Use algorithm knowledge with loop statements using Python.
- 12.4.1f:** Create strings using Python.
- 12.4.1g:** Construct list data types with Python.
- 12.4.1h:** Create tuples using Python.
- 12.4.1i:** Access dictionary elements using keys.
- 12.4.1j:** Manipulating date and time elements with Python.
- 12.4.1k:** Create a function using Python.
- 12.4.1l:** Produce I/O files using Python.
- 12.4.1m:** Use exception handling and/or assertion to handle any unexpected error in a Python program.

Content Expansion

Body of Knowledge for Understanding

- Introduction to Python⁴
 - What is Python Programming Language
 - High-Level Language
 - General Purpose
 - Interpreted
 - Why Python
 - Python Versions
- Environment Setup of Python
 - Install Python on Windows
 - Verify Installation
- Setup Visual Studio Code for Python
 - Setting up
 - Installing Python Extension
- Develop the “Hello”World” Program with Python.
 - Creating a new Python Project
 - What is a function
 - Executing the Python Hello World Program
 - Python IDLE
- Syntax of Python
 - White-space and Indentation
 - Comments
 - Continuation of Statements
 - Identifiers

⁴ [Python Tutorial - Learn Python Programming from Scratch](#)

- Keywords
 - String Literals
- Variable Types of Python
 - What is a Variable?
 - Creating Variables
 - Naming Variables
- Python Strings
 - Introduction to Python String
 - Creating Multi-line string
 - Using variables in Python strings with the f-string
 - Concatenating Python Strings
 - Accessing String Elements
 - Getting the Length of a string
 - Slicing strings
 - Python strings are immutable
- Python Number
 - Integers
 - Floats
 - Underscores in Numbers
- Python Boolean
 - Introduction to Python Boolean Data Type
 - The bool() function
 - Falsy and Truthy values
- Python Constants
- Add Comments to Code using Python
 - Introduction
 - Python Block Comments
 - Python Online Comments
 - Python Inline Comments
 - Python docstrings
 - One-line
 - Multi-line
- Type Conversion in Python
 - Other Type Conversion Functions
 - Getting the Type of Value
- Basic Operations of Python
 - Introduction to Comparison Operators
 - Less than operator (<)
 - Less than or equal to operator (<=)
 - Greater than operator (>)
 - Greater than or equal to operator (>=)
 - Equal To Operator (==)
 - Not Equal To Operator (!=)
 - Introduction to Logical Operators
 - The and operator
 - The or operator
 - The not operator

- Precedence and Logical Operators
- Python Control Flow
 - If Statement
 - If...else statement
 - If...elif...else...statement
 - Introduction to Ternary Operator
 - Introduction to Python for Loop Statement with the range() function
 - Specifying the start value of a sequence
 - Specifying the increment of a sequence
 - Calculate the sum of a sequence with Python Loop
 - Introduction to Python while Statement
 - Examples
 - How it works
 - Introduction to Python break Statement
 - Using break with for loop
 - Using break with a while loop
 - Introduction to Python continue statement
 - Using continue in a for loop example
 - Using python in a while loop example
 - Introduction to Python pass Statement
 - Use with if statement
 - Use with for statement
 - Use with while statement
 - Use with functions and classes
- Python Functions
 - What is Function
 - Why do we need Functions
 - Define Python Function
 - Function Definition
 - Function Body
 - Calling a function
 - Passing information to Python Functions
 - Return Value
 - Python functions with multi parameters
 - Introduction to Python default parameters
 - Example
 - Multiple default parameters
 - Introduction to Python Keyword Arguments
 - Keyword Arguments and Default Parameters
 - Python Keyword Argument Requirements
 - Introduction to Python Recursive Functions
 - Example
 - What are Python Lambda expression
 - Examples
 - Introduction to help() function
 - Using docstrings to document functions
- Python Lists

- What is a list?
- Accessing elements in a list
- Modifying, adding, and removing elements
- Introduction to Python tuples
 - Define Tuple
 - Define Tuple with one element
 - Assigning a tuple
- Introduction to Python List sort() method.
 - Examples
 - Using lambda expression
- Introduction to Python sorted() function
 - Examples
- Introduction to Python List slice notation
 - Examples
- Introduction to List Unpacking
 - Unpacking and Packing
- Python iterate over a List
 - Use Python for Loop to iterate over list with index
- How to Find the Index of an Element in a List
- Introduction to Python Iterables
 - What is an Iterator
- Introduction to Python map() function
 - Examples
- Introduction to Python filter() function
- Reducing a List
 - Python reduce() function
- Introduction to Python List Comprehension
 - With if condition
- Python Dictionary
 - Introduction to Python Dictionary Type
 - Access values in Dictionary
 - Adding new key-value pairs
 - Modifying vales in a key-value pair
 - Removing key-value pair
 - Looping through a dictionary
- Python Dictionary Comprehension
 - Examples
- Python Set Type
 - Introduction
 - Getting sizes of a set
 - Checking an element in a set
 - Adding elements to set
 - Removing elements from set
 - Returning element from set
 - Frozen a set
 - Set comprehension
 - Examples
 - Set Union

- Union() method
 - | operator
 - Union() method vs. Set union operator
- Set Interaction
 - Introduction
 - Set intersection() method vs. St intersection operator (&)
- Set Differences
 - Set difference() method vs. Set difference operator (-)
- Symmetric difference of sets
 - Symmetric_difference() method vs. Symmetric difference operator (^)
- Python issubset() Method
 - Subset operators
- Introduction to Python isupersest
 - Issuperset() method
 - Superset operators
- Python disjoint sets
 - Isdisjoint() method

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Topic 2: Advance Python

Performance Indicators

- 12.4.2a:** Create a class of objects using Python.
- 12.4.2b:** Creating Instance Objects.
- 12.4.2c:** Accessing object attributes.
- 12.4.2d:** Destroying Objects.
- 12.4.2e:** Creating a CGI program.
- 12.4.2f:** Creating a database and interfacing your program to the database.
- 12.4.2g:** Utilize socket programming using Python.
- 12.4.2h:** Creating a multithread program.
- 12.4.2i:** Build a software application for your school.

Content Expansion

Body of Knowledge for Understanding

- Exception Handling
 - Python try...except statement
 - Syntax errors
 - Exceptions
 - Handling exceptions
 - Catching specific exceptions
 - Handling multiple exceptions
 - Python try...catch...finally statement
 - Examples
 - Python try...except...else statement
 - Examples
- More on Python Loops
 - Python for else statement
 - Examples
 - Python while else statement
 - Example
 - Python do...while loop statement
- More on Python Functions
 - Python Unpacking Tuple
 - Unpacking a tuple
 - Unpacking tuple to swap values of two variables
 - ValueError: too many values to unpack
 - Extended unpacking using the * operator
 - Using the * operator on the right hand side
 - Python *args
 - *args Parameter
 - *args argument exhaust positional arguments
 - Unpacking arguments
 - **kwargs parameters
 - * args and **kwargs arguments
 - Python Partial Functions
- Python Modules and Packages
 - Python Modules
 - Writing Python Modules
 - Importing Python Module Objects
 - Python Module Search Path
 - Python _name_

- Python packages
- Working with Files
 - Python Read Text File
 - TL;DR
 - Steps for Reading a Python Text File
 - Examples
 - Read UTF-8 text files
 - Python Write Text File
 - TL;DR
 - Steps for Writing to Text files
 - Examples
 - Appending Text files
 - Writing to a UTF-8 text file
 - Python Create Text File
 - Open() to create text file
 - Python Check if file exists
 - Os.path.exists() function
 - Path.is_file() method
 - Python Read CSV file
 - Python Write CSV File
- Third-Party Packages, PIP and Virtual Environment
 - Python Package Index (PyPI)
 - Package version
 - Pip
 - Install package
 - Listed installed package
 - Uninstall package
 - Dependencies of package
 - Virtual Environments
 - Install pipenv Windows
 - Python pipenv

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